# HWRS 561b: Physical Hydrogeology II

## Steady-state 1D unsat flow (Part 1)

Agenda: Steady-state unsaturated flow
Reading: Chapter 11 (Pinder & Celia, 2006)

# Review

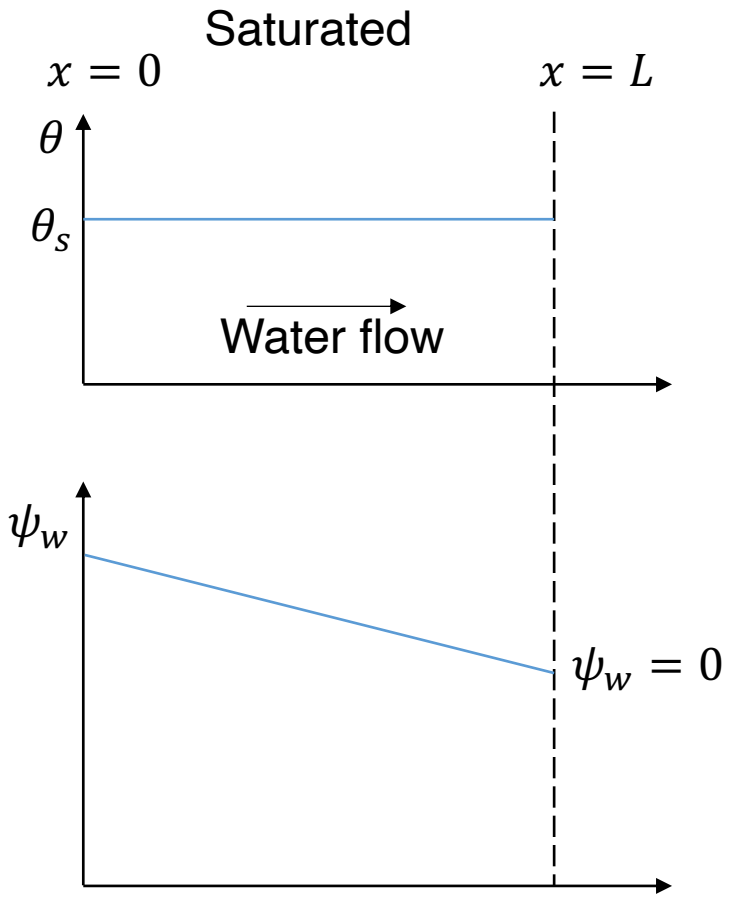


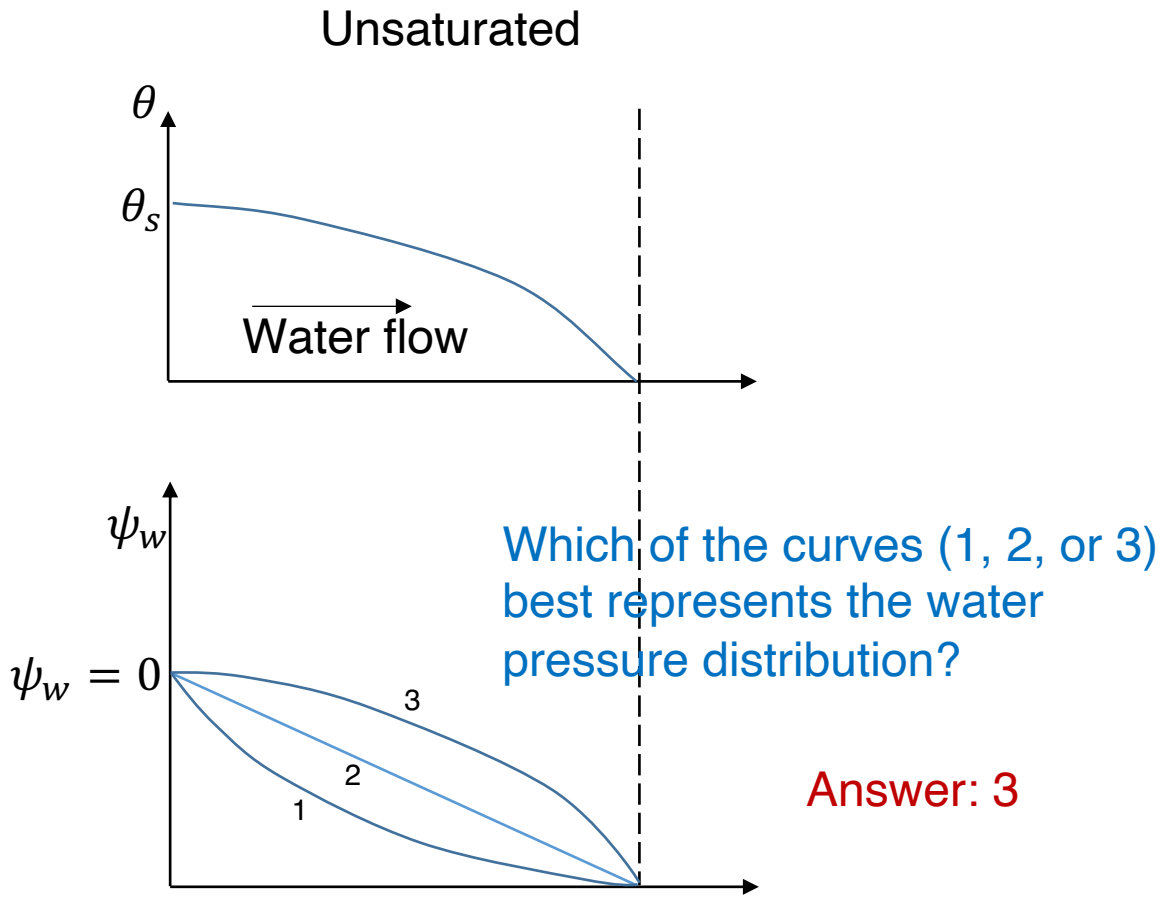
- Three forms of Richards' equation
  - Mixed form
  - Pressure head-based form
    - ✓ Specific moisture capacity
  - Water content-based form
    - ✓ Soil moisture diffusivity (What does the equation have to do with "diffusion"?)
    - ✓ Cannot be used if the domain involves saturated water flow
  - How to include soil and fluid compressibility?
- Richards' assumptions
  - Air pressure remains almost zero everywhere, but air does move.
  - Does "air movement" make the Richards' equation invalid? No, as long as air pressure remains almost zero everywhere.

# Steady-State Unsaturated Flow



Horizontal flow (Homogeneous column)




Which of the curves (1, 2, or 3) best represents the water pressure distribution?

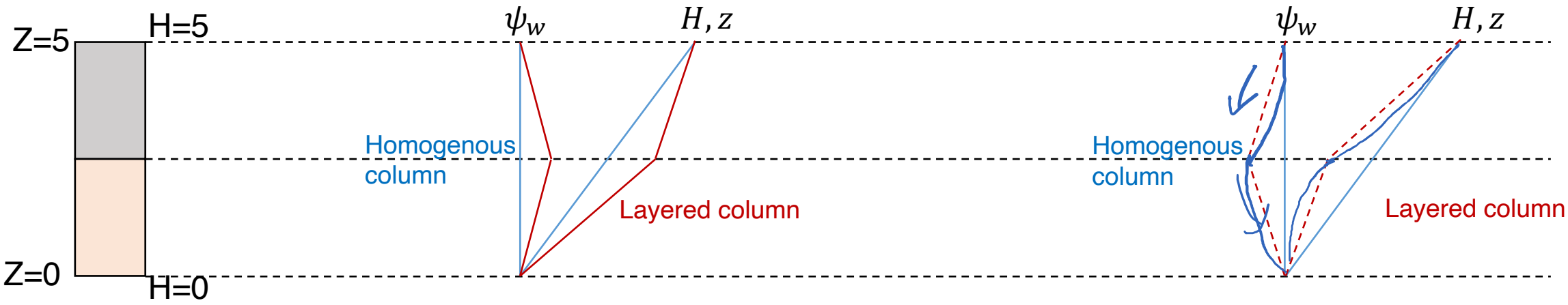
Answer: 3

- Unsaturated flow involves nonlinearities that make their behaviors differ from that of the saturated flow

# Steady-State Unsaturated Flow



Vertical flow (in layered columns)




Coarse over fine

Fine over coarse

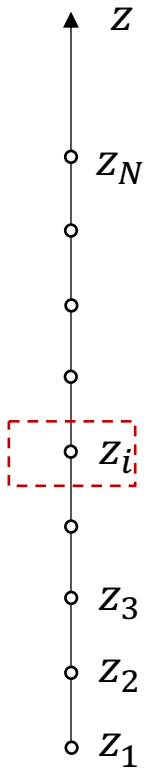
- Hydraulic head and pressure head are both continuous in space.
- Is water saturation continuous?

# Steady-State Unsaturated Flow: Numerical Soln.

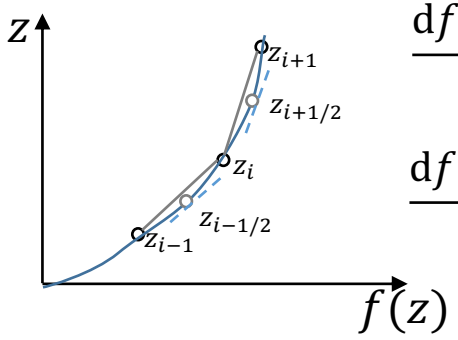


$$\frac{\partial \theta_w}{\partial t}^{0} - \frac{\partial}{\partial z}\left(K\frac{\partial \psi_w}{\partial z}\right) - \frac{\partial K}{\partial z} = 0 \quad \text{Steady-state} \Rightarrow \quad \underbrace{\frac{\mathrm{d}}{\mathrm{d}z}\left(K\frac{\mathrm{d}\psi_w}{\mathrm{d}z}\right)}_{①} + \underbrace{\frac{\mathrm{d}K}{\mathrm{d}z}}_{②} = 0 \qquad (1)$$

- Equation (1) is a second-order ordinary differential equation in 1D.
- It is nonlinear because $K = K(\psi_w)$ is a nonlinear function.
- To solve it, we need <u>two boundary conditions</u> and we need to do it in an <u>iterative</u> procedure.




- How to solve this 1D **nonlinear ordinary differential equation**?
- <u>Key idea</u>: Divide the domain into many boxes and convert the **differential equation** to a system of nonlinear **algebraic equations**.
- Technique: Use finite difference to approximate derivatives



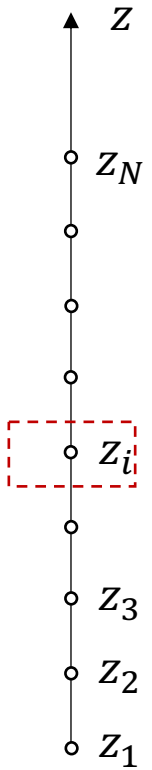

$$\frac{\mathrm{d}f(z_{i+1/2})}{\mathrm{d}z} \approx \frac{f(z_{i+1}) - f(z_i)}{z_{i+1} - z_i}$$

$$\frac{\mathrm{d}f(z_{i-1/2})}{\mathrm{d}z} \approx \frac{f(z_i) - f(z_{i-1})}{z_i - z_{i-1}}$$

# Steady-State Unsaturated Flow: Numerical Soln.



$$\frac{\partial \theta_w}{\partial t}^{0} - \frac{\partial}{\partial z}\left(K\frac{\partial \psi_w}{\partial z}\right) - \frac{\partial K}{\partial z} = 0 \quad \text{Steady-state} \Rightarrow \quad \underbrace{\frac{\mathrm{d}}{\mathrm{d}z}\left(K\frac{\mathrm{d}\psi_w}{\mathrm{d}z}\right)}_{①} + \underbrace{\frac{\mathrm{d}K}{\mathrm{d}z}}_{②} = 0 \qquad (1)$$




$$① \quad \left.\frac{\mathrm{d}}{\mathrm{d}z}\left(K\frac{\mathrm{d}\psi_w}{\mathrm{d}z}\right)\right|_{z_i} \approx \frac{\left(K\frac{\mathrm{d}\psi_w}{\mathrm{d}z}\right)_{i+1/2} - \left(K\frac{\mathrm{d}\psi_w}{\mathrm{d}z}\right)_{i-1/2}}{\Delta z} \approx \frac{K_{i+1/2}\frac{\psi_{w,i+1}-\psi_{w,i}}{\Delta z} - K_{i-1/2}\frac{\psi_{w,i}-\psi_{w,i-1}}{\Delta z}}{\Delta z}$$

$$= \frac{K_{i+1/2}(\psi_{w,i+1}-\psi_{w,i}) - K_{i-1/2}(\psi_{w,i}-\psi_{w,i-1})}{\Delta z^2}$$

$$② \quad \left.\frac{\mathrm{d}K}{\mathrm{d}z}\right|_{z_i} \approx \frac{K_{i+1/2}-K_{i-1/2}}{\Delta z}$$

$$\Rightarrow \frac{K_{i+1/2}}{\Delta z^2}(\psi_{w,i+1}-\psi_{w,i}) - \frac{K_{i-1/2}}{\Delta z^2}(\psi_{w,i}-\psi_{w,i-1}) + K_{i+1/2}\frac{1}{\Delta z} - K_{i-1/2}\frac{1}{\Delta z} = 0$$

$$\Rightarrow \frac{K_{i+1/2}}{\Delta z^2}\psi_{w,i+1} - \left(\frac{K_{i+1/2}}{\Delta z^2} + \frac{K_{i-1/2}}{\Delta z^2}\right)\psi_{w,i} + \frac{K_{i-1/2}}{\Delta z^2}\psi_{w,i-1} + K_{i+1/2}\frac{1}{\Delta z} - K_{i-1/2}\frac{1}{\Delta z} = 0$$

This is an algebraic equation with 3 unknowns. We can write such an algebraic equation for each node or box and we can get $N$ algebraic equations for the $N$ unknowns $(\psi_{w,1}, \psi_{w,2}, \ldots, \psi_{w,N})$

# Steady-State Unsaturated Flow: Numerical Soln.



In matrix form:

$$\begin{pmatrix} 1 & 0 & 0 & & & & & \\ \frac{K_{3/2}}{\Delta z^2} & -\frac{(K_{3/2}+K_{5/2})}{\Delta z^2} & \frac{K_{5/2}}{\Delta z^2} & & & & & \\ & \frac{K_{5/2}}{\Delta z^2} & -\frac{(K_{5/2}+K_{7/2})}{\Delta z^2} & \frac{K_{7/2}}{\Delta z^2} & & & & \\ & & \ddots & \ddots & \ddots & & & \\ & & & \frac{K_{i-1/2}}{\Delta z^2} & -\frac{(K_{i-1/2}+K_{i+1/2})}{\Delta z^2} & \frac{K_{i+1/2}}{\Delta z^2} & & \\ & & & & \ddots & \ddots & \ddots & \\ & & & & & \frac{K_{n-1-1/2}}{\Delta z^2} & -\frac{(K_{n-1-1/2}+K_{n-1+1/2})}{\Delta z^2} & \frac{K_{n-1+1/2}}{\Delta z^2} \\ & & & & & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \psi_{w,1} \\ \psi_{w,2} \\ \psi_{w,3} \\ \vdots \\ \psi_{w,i} \\ \vdots \\ \psi_{w,n-1} \\ \psi_{w,n} \end{pmatrix} = \begin{pmatrix} B.C. \\ \frac{K_{3/2}-K_{5/2}}{\Delta z} \\ \frac{K_{5/2}-K_{7/2}}{\Delta z} \\ \vdots \\ \frac{K_{i-1/2}-K_{i+1/2}}{\Delta z} \\ \vdots \\ \frac{K_{n-1-1/2}-K_{n-1+1/2}}{\Delta z} \\ B.C. \end{pmatrix}$$

$\Rightarrow \boldsymbol{F}(\boldsymbol{\psi_w}) \cdot \boldsymbol{\psi_w} = \boldsymbol{R}$ How to solve this nonlinear system of algebraic equations?

=> Employ an iterative methods

# Steady-State Unsaturated Flow: Numerical Soln.



$$\frac{\partial \theta_w}{\partial t} + \frac{\partial}{\partial z}\left(-K\left(\frac{\partial \psi_w}{\partial z} + 1\right)\right) = 0 \qquad q_z = -K\left(\frac{\partial \psi_w}{\partial z} + 1\right)$$

Assumes $z$ positive upward

$$\frac{\partial \theta_w}{\partial t} + \frac{\partial}{\partial z}\left(-K\left(\frac{\partial \psi_w}{\partial z} - 1\right)\right) = 0 \qquad q_z = -K\left(\frac{\partial \psi_w}{\partial z} - 1\right)$$

Assumes $z$ positive downward

Which equation is correct?

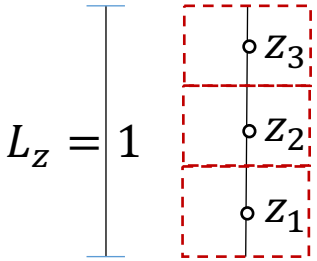
Now, we assume $z$ is positive upward.
Which equation is correct?

# Steady-State Unsaturated Flow: Numerical Soln.



$$\frac{\mathrm{d}}{\mathrm{d}z}\left(K\frac{\mathrm{d}\psi_w}{\mathrm{d}z}\right)+\frac{\mathrm{d}K}{\mathrm{d}z}=0$$

$\psi_w(z_3)=2$




$\psi_w(z_1)=0$

$\Delta z = z_2 - z_1$
$= z_3 - z_2 = 1/3$

Equation for box ①:

$$\psi_{w,1}=0$$

Equation for box ③:

$$\psi_{w,3}=2$$

Equation for box ②:

$$\frac{\left(K\frac{d\psi_w}{dz}\right)\Big|_{z+\frac{1}{2}}-\left(K\frac{d\psi_w}{dz}\right)_{z-\frac{1}{2}}}{\Delta z}+\frac{K_{z+\frac{1}{2}}-K_{z-\frac{1}{2}}}{\Delta z}=0$$

$$\Rightarrow \frac{K_{z+\frac{1}{2}}\frac{\psi_{w,3}-\psi_{w,2}}{\Delta z}-K_{z-\frac{1}{2}}\frac{\psi_{w,2}-\psi_{w,1}}{\Delta z}}{\Delta z}+\frac{K_{z+\frac{1}{2}}-K_{z-\frac{1}{2}}}{\Delta z}=0$$

$$\Rightarrow \frac{K_{z+\frac{1}{2}}}{\Delta z^2}\psi_{w,3}+\left(-\frac{K_{z+\frac{1}{2}}}{\Delta z^2}+\frac{K_{z-\frac{1}{2}}}{\Delta z^2}\right)\psi_{w,2}+\frac{K_{z-\frac{1}{2}}}{\Delta z^2}\psi_{w,1}=-\frac{K_{z+\frac{1}{2}}-K_{z-\frac{1}{2}}}{\Delta z}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ \frac{K_{z-\frac{1}{2}}}{\Delta z^2} & -\frac{K_{z+\frac{1}{2}}}{\Delta z^2}+\frac{K_{z-\frac{1}{2}}}{\Delta z^2} & \frac{K_{z+\frac{1}{2}}}{\Delta z^2} \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} \psi_{w,1} \\ \psi_{w,2} \\ \psi_{w,3} \end{bmatrix}=\begin{bmatrix} 0 \\ -\frac{K_{z+\frac{1}{2}}-K_{z-\frac{1}{2}}}{\Delta z} \\ 2 \end{bmatrix}$$

For our given B.C.:

$$\psi_{w,2}=1$$

⇒ Everywhere is saturated!

For another B.C.:

e.g.: $\psi_{w,1}=-3$

$\psi_{w,3}=-1$

need to solve iteratively